

Manas Tripathi

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Education

Indian Institute of Information Technology, Ranchi

Aug 2023 – present

BTech in Computer Science and Engineering

- CGPA: 9.30/10.0

Experience

Summer Research Intern — Indian Institute of Technology, Gandhinagar

May 2025 – Jul 2025

- Designed and implemented a real-time bias correction framework for Ensemble Rainfall Forecasts (ERFS) under **Prof. Vimal Mishra**, leveraging **LSTM** and **Knowledge Distillation** to improve forecast accuracy and model generalization.
- Enhanced forecasting performance by achieving **+0.57 NSE**, reducing **MAE by 32.7%**, and lowering **PBIAS by 18%** over baseline models, quantifiably strengthening flood prediction reliability across climate-vulnerable river basins and supporting decision-making for **2M+ farmers**.

Projects

Verge

[Live](#) | [GitHub](#)

- Built and deployed an end-to-end **MLOps pipeline** for GPU and CPU energy consumption prediction, trained on **10K+ hardware-software profiles** to achieve **$R^2 = 0.89$** and **12.3% MAPE**, outperforming baseline models.
- Containerized the application with **Docker** and deployed it as a scalable **FastAPI** on **Kubernetes**, serving **25+ requests/second** with **107 ms latency** under **1,170 concurrent users**.
- Automated **CI/CD** and ML workflows using **MLflow**, **DVC**, **Prefect**, and **GitHub Actions**, reducing deployment time by **60%** and retraining time by **40%**.

MindScope

[GitHub](#)

- Fine-tuned a **BERT** transformer using **transfer learning** on **15K+ Reddit mental health posts** for stress and anxiety classification, achieving **65% accuracy**, **0.72 F1-score**, and **100 ms CPU inference latency**.
- Optimized classification thresholds using **ROC analysis**, improving precision to **78%** and recall to **68%**, while enabling scalable inference at **50+ requests/second**.

EmotionGAN

[GitHub](#)

- Pre-trained a **VGG16** model using **transfer learning** for facial emotion recognition, achieving **58% accuracy** across **7 emotion classes**.
- Analyzed artwork emotions using the trained model and developed a **Conditional GAN (cGAN)** to generate emotion-aligned images, combining **computer vision**, **deep learning**, and **generative AI**.

Voicemote

[GitHub](#)

- Developed a speech emotion recognition model using **Librosa**, extracting **MFCCs**, **chroma**, and **spectral** features from raw audio for emotion classification.
- Achieved **72% training accuracy**, validating the effectiveness of handcrafted audio features for **human-computer interaction** and **emotion-aware AI** applications.

Technical Skills

Languages: Python, C++, C, SQL, JavaScript

ML: Machine Learning, Deep Learning, Reinforcement Learning, Transformers, LLMs, VLMs, CNNs, LSTMs, GANs, VAEs, Transfer Learning, Fine-Tuning, Knowledge Distillation, Computer Vision, NLP, Time Series Forecasting, A/B Testing

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, Scikit-learn, XGBoost, LightGBM, Keras, OpenCV, Librosa, Pandas, NumPy, Matplotlib, React

LLM & Software Engineering: LangChain, RAG, Vector Databases, AI Agents, FastAPI, Node.js, Express.js

MLOps & Deployment: Docker, Kubernetes, CUDA, MLflow, DVC, Prefect, Git, GitHub Actions, CI/CD

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Graph Theory

Achievements

- Secured **AIR 6499** in **GATE CS 2026**, ranking in the **top 2.5%** among **250K+ candidates** nationwide.
- Finalist in **Smart India Hackathon 2024**, developing an **AI-powered job seeker-recruiter platform** under the **Government of Rajasthan's** Smart Education initiative.